Amir Beck, Chapter 1 - Exercise 1.8

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1.8. Suppose that $\mathbb{R}^n$ and $\mathbb{R}^m$ are equipped with norms $||.||_b$ and $||.||_a$, respectively. Show that the induced norm $||.||_{a,b}$ can be computed by the formula

$$||\mathbf{A}||_{a,b} = \max_{\mathbf{x}}\{||\mathbf{Ax}||_{\mathbf{b}} : ||\mathbf{x}||_{\mathbf{a}} = \mathbf{1}\}.$$

Recall that $||\mathbf{A}||_{a,b} := \max_{\mathbf{x}}\{||\mathbf{Ax}||_b : ||\mathbf{x}||_a \leq 1\}$

Let $C = \{\mathbf{x} \in \mathbb{R}^n : ||\mathbf{x}||_a \leq 1\}$ and $C' = \{\mathbf{x} \in \mathbb{R}^n : ||\mathbf{x}||_a = 1\}$. Obviously, $C' \subseteq C$, then

$$\max_{\mathbf{x}}\{||\mathbf{Ax}||_b : ||\mathbf{x}||_a \leq 1\} \geq \max_{\mathbf{x}}\{||\mathbf{Ax}||_b : ||\mathbf{x}||_a = 1\},$$

that is,

$$||\mathbf{A}||_{a,b} \geq \max_{\mathbf{x}}\{||\mathbf{Ax}||_b : ||\mathbf{x}||_a = 1\}.$$

Let $\mathbf{x} \in \mathbb{R}^n$ such that $0 < ||\mathbf{x}||_a \leq 1$. Consider $\mathbf{y} = \frac{\mathbf{x}}{||\mathbf{x}||_a}$. Note that $||\mathbf{y}||_a = \frac{||\mathbf{x}||_a}{||\mathbf{x}||_a} = 1$, so $\mathbf{y} \in C'$. Furthermore

$$||\mathbf{Ay}||_b = \left|\left|\mathbf{A}\frac{\mathbf{x}}{||\mathbf{x}||_{\mathbf{a}}}\right|\right|_b = ||\mathbf{Ax}||_b \frac{1}{||\mathbf{x}||_a}$$

$$\Leftrightarrow$$

$$||\mathbf{Ax}||_b \leq ||\mathbf{Ay}||_b ||\mathbf{x}||_a \leq ||\mathbf{Ay}||_b.$$

That is, for every $\mathbf{x} \in C \setminus \{\mathbf{0}\}$, we constructed a $\mathbf{y} \in C'$ such that $||\mathbf{Ax}||_b \leq |\mathbf{Ay}||_b$. For $\mathbf{x} = \mathbf{0}$, we can take any $\mathbf{y} \in C'$, because $||\mathbf{Ay}||_b \geq 0 = ||\mathbf{A}.\mathbf{0}||_b$. Therefore,

$$||\mathbf{A}||_{a,b} \leq \max_{\mathbf{x}}\{||\mathbf{Ax}||_b : ||\mathbf{x}||_a = 1\}.$$